the relationship set *borrower* is total. In contrast, an individual can be a bank customer whether or not she has a loan with the bank. Hence, it is possible that only some of the *customer* entities are related to the *loan* entity set through the *borrower* relationship, and the participation of *customer* in the *borrower* relationship set is therefore partial.

2.3 Keys

We must have a way to specify how entities within a given entity set are distinguished. Conceptually, individual entities are distinct; from a database perspective, however, the difference among them must be expressed in terms of their attributes.

Therefore, the values of the attribute values of an entity must be such that they can *uniquely identify* the entity. In other words, no two entities in an entity set are allowed to have exactly the same value for all attributes.

A *key* allows us to identify a set of attributes that suffice to distinguish entities from each other. Keys also help uniquely identify relationships, and thus distinguish relationships from each other.

2.3.1 Entity Sets

A **superkey** is a set of one or more attributes that, taken collectively, allow us to identify uniquely an entity in the entity set. For example, the *customer-id* attribute of the entity set *customer* is sufficient to distinguish one *customer* entity from another. Thus, *customer-id* is a superkey. Similarly, the combination of *customer-name* and *customer-id* is a superkey for the entity set *customer*. The *customer-name* attribute of *customer* is not a superkey, because several people might have the same name.

The concept of a superkey is not sufficient for our purposes, since, as we saw, a superkey may contain extraneous attributes. If K is a superkey, then so is any superset of K. We are often interested in superkeys for which no proper subset is a superkey. Such minimal superkeys are called **candidate keys**.

It is possible that several distinct sets of attributes could serve as a candidate key. Suppose that a combination of *customer-name* and *customer-street* is sufficient to distinguish among members of the *customer* entity set. Then, both {*customer-id*} and {*customer-name*, *customer-street*} are candidate keys. Although the attributes *customer-id* and *customer-name* together can distinguish *customer* entities, their combination does not form a candidate key, since the attribute *customer-id* alone is a candidate key.

We shall use the term **primary key** to denote a candidate key that is chosen by the database designer as the principal means of identifying entities within an entity set. A key (primary, candidate, and super) is a property of the entity set, rather than of the individual entities. Any two individual entities in the set are prohibited from having the same value on the key attributes at the same time. The designation of a key represents a constraint in the real-world enterprise being modeled.

Candidate keys must be chosen with care. As we noted, the name of a person is obviously not sufficient, because there may be many people with the same name. In the United States, the social-security number attribute of a person would be a

candidate key. Since non-U.S. residents usually do not have social-security numbers, international enterprises must generate their own unique identifiers. An alternative is to use some unique combination of other attributes as a key.

The primary key should be chosen such that its attributes are never, or very rarely, changed. For instance, the address field of a person should not be part of the primary key, since it is likely to change. Social-security numbers, on the other hand, are guaranteed to never change. Unique identifiers generated by enterprises generally do not change, except if two enterprises merge; in such a case the same identifier may have been issued by both enterprises, and a reallocation of identifiers may be required to make sure they are unique.

2.3.2 Relationship Sets

The primary key of an entity set allows us to distinguish among the various entities of the set. We need a similar mechanism to distinguish among the various relationships of a relationship set.

Let *R* be a relationship set involving entity sets $E_1, E_2, \ldots, E_n$. Let *primary-key*(E_i) denote the set of attributes that forms the primary key for entity set E_i. Assume for now that the attribute names of all primary keys are unique, and each entity set participates only once in the relationship. The composition of the primary key for a relationship set depends on the set of attributes associated with the relationship set *R*.

If the relationship set *R* has no attributes associated with it, then the set of attributes

$$\textit{primary-key}(E_1) \cup \textit{primary-key}(E_2) \cup \cdots \cup \textit{primary-key}(E_n)$$

describes an individual relationship in set R.

If the relationship set *R* has attributes $a_1, a_2, \cdots, a_m$ associated with it, then the set of attributes

$$\textit{primary-key}(E_1) \cup \textit{primary-key}(E_2) \cup \cdots \cup \textit{primary-key}(E_n) \cup \{a_1, a_2, \ldots, a_m\}$$

describes an individual relationship in set R.

In both of the above cases, the set of attributes

$$\textit{primary-key}(E_1) \cup \textit{primary-key}(E_2) \cup \cdots \cup \textit{primary-key}(E_n)$$

forms a superkey for the relationship set.

In case the attribute names of primary keys are not unique across entity sets, the attributes are renamed to distinguish them; the name of the entity set combined with the name of the attribute would form a unique name. In case an entity set participates more than once in a relationship set (as in the *works-for* relationship in Section 2.1.2), the role name is used instead of the name of the entity set, to form a unique attribute name.